

drawing the line

digitalisierung und programmierung · input, session 4

agenda

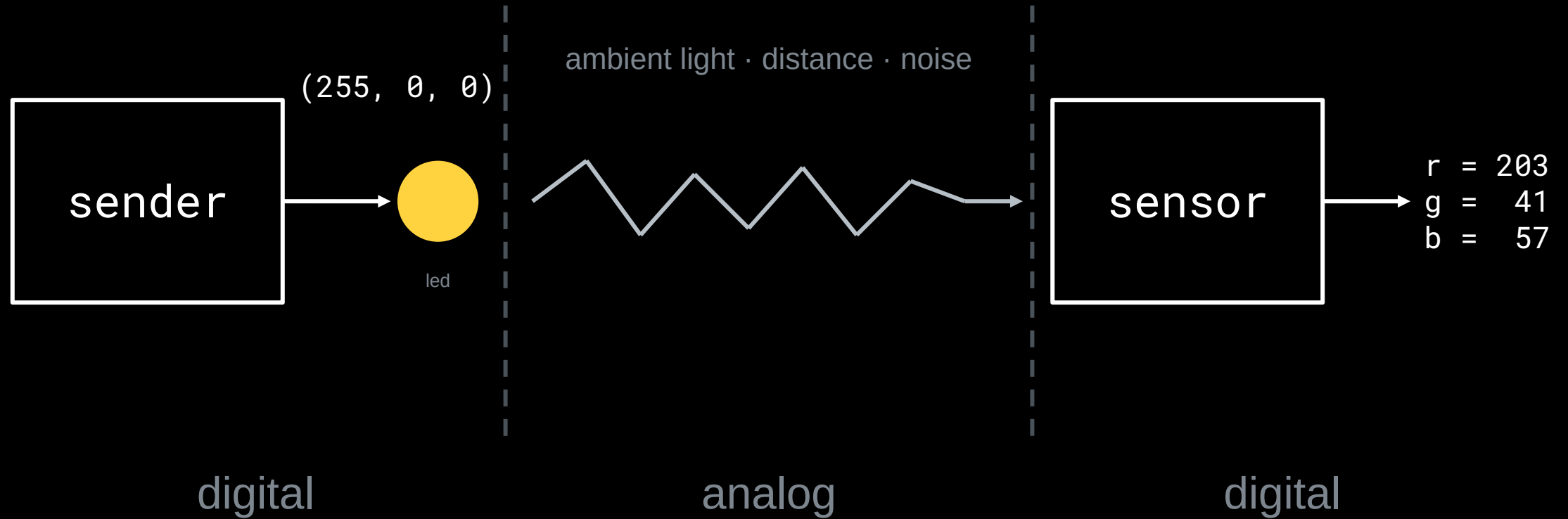
- 1 a world without steps
- 2 drawing your own lines

a world without steps

is this
digital?



the part no program controls



the middle zone

light is analog: between any two
brightnesses lies another.

the middle zone

light is analog: between any two
brightnesses lies another.

your program needs states:
finitely many, clearly apart.

digital is not the same as electronic

digital

split-flap board

light switch

abacus

a die

analog

slide rule

dial thermometer

dimmer

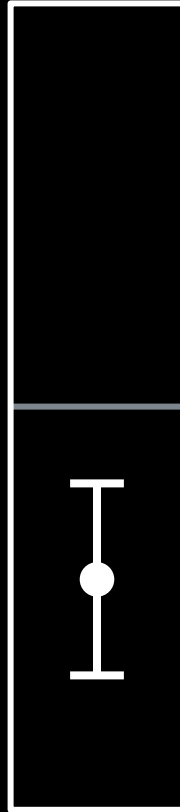
hourglass

not one of them contains a chip.

digital means finitely many states,
and nothing in between counts.

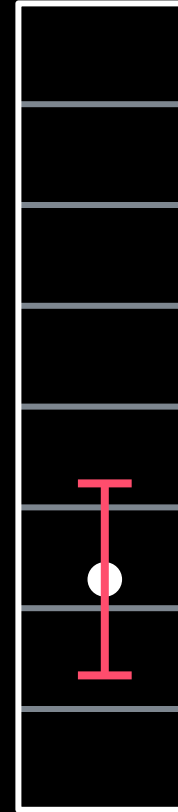
same noise, more regions

two regions



one measurement
with its wobble

eight regions

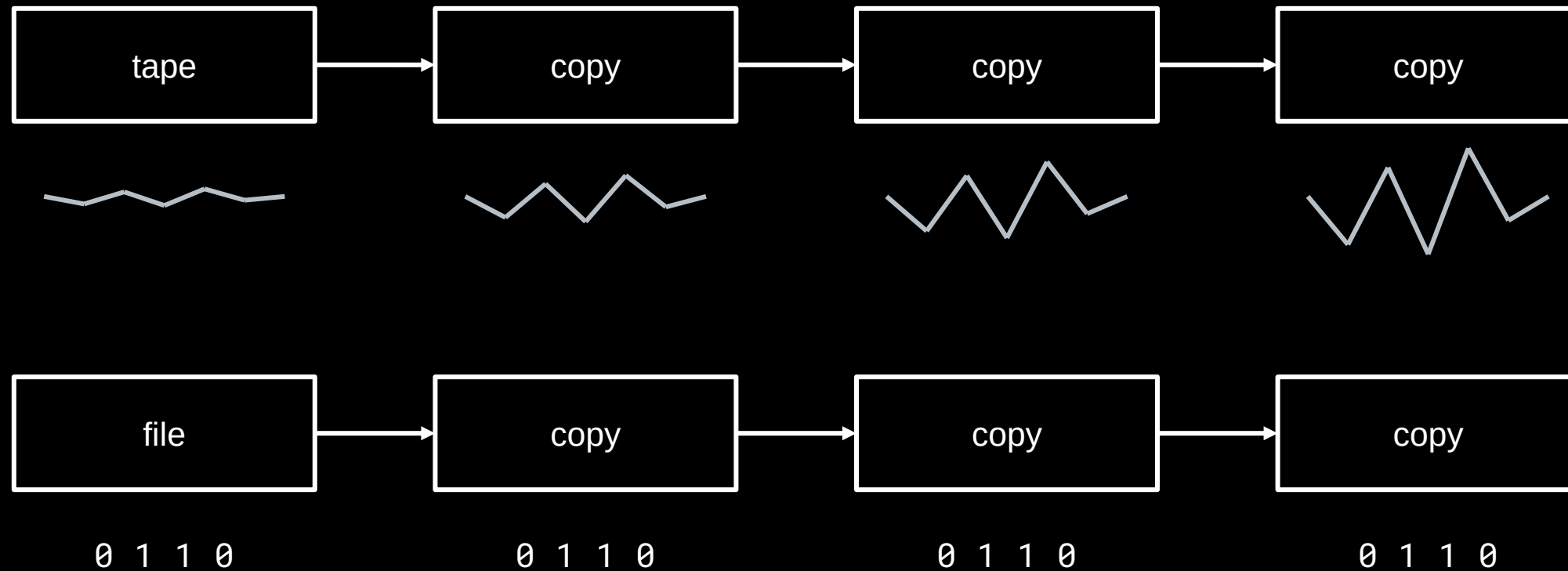


the noise did not grow. the margin shrank.

digitising throws information away.
on purpose.

which exact value it was inside the region: gone for good.

what the thrown-away precision buys



every digital copy is a rebirth from clean states.

drawing your own lines

where your alphabet is born

```
value = sensor.get_clear()  
if value > 250:  
    symbol = "bright"  
else:  
    symbol = "dark"
```

the sensor delivers hundreds of possible numbers.
this line turns them into two symbols.

how many colours are safe? your measurements decide.

two are safe. eight are fast. somewhere in between sits your alphabet, and only your own data knows where.